

AI-Powered Code Review Web Application Prompt

Objective:

Build a full-featured AI-powered Code Review Web Application for multiple programming languages: JavaScript, Python, React, TypeScript, C, C++, PHP. The app must be built with Next.js 15 (App Router), Tailwind CSS, and follow the Atomic Design Pattern for reusable UI components. The app should provide developers with real-time AI code reviews, suggestions, fixed code, and security warnings.

1. Core Features

1.1 Authentication System

- NextAuth.js authentication via Email, Google, GitHub.
- Personalized dashboard for authenticated users.
- Route protection for unauthenticated access.

1.2 Dashboard

- Sidebar navigation: Home, New Review, History, Profile, Settings.
- Main section:
 - Language Selector: JS, Python, React, TS, C, C++, PHP.
 - Code Editor: Monaco Editor or CodeMirror with syntax highlighting.
 - File Upload: Accepts .js, .py, .ts, .cpp, .c, .php files.
 - Buttons: Submit for Review, Upload File.
 - AI Review Output Tabs: Suggestions, Fixed Code, Security Warnings.
 - Loading state during AI processing.

1.3 Review History

- Table/grid listing past reviews.
- Columns: Date, Language, File/Code Name, Status, Action (View Report/Reopen).
- Pagination and search by date, language, or file.

1.4 Profile Page

- User info: Name, Email, Plan (Free/Pro/Enterprise).
- API usage stats and plan management.

1.5 Landing Page (Public)

- Hero Section: "AI-Powered Code Reviews Instantly".
- Call-to-action buttons: Get Started, See Demo.
- Features Section: Multi-language support, inline suggestions, security detection.
- Footer: Docs, Pricing, Contact.

1.6 Optional Admin Panel

- User management and subscription overview.

- Usage analytics and AI model performance monitoring.

2. Technical Stack

Frontend: Next.js 15 (App Router), Tailwind CSS, Framer Motion.

Editor: Monaco Editor (@monaco-editor/react).

Backend/API: Next.js API Routes, AI integration (OpenAI/Claude/custom model).

Database: PostgreSQL or MongoDB.

Auth: NextAuth.js (GitHub, Google, Email).

Optional: Redux Toolkit or Zustand for state management, Toast notifications.

3. Development Roadmap

Phase 1 (MVP): Authentication, dashboard, code editor, AI review output.

Phase 2: Review history table, export/download fixed code, optimize editor.

Phase 3: Admin Panel, collaborative reviews, advanced AI insights.

4. Deliverables

- Next.js 15 project with Atomic Design structure.
- Tailwind CSS styling for all components.
- Fully functional AI integration.
- Responsive design (desktop & mobile).
- Clear documentation for setup, AI integration, and future enhancements.

Output-Oriented Notes:

- Provide code snippets for authentication, dashboard, Monaco editor setup, AI API integration.
- Tailwind CSS responsive layouts for dashboard, review table, landing page.
- Folder structure and component organization (Atomic Design).
- Error handling and loading states for AI processing.
- Best practices for security, file handling, API usage limits.